Project: Worked examples
Title: Plane wall heat transfer (layers and films)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Steady heat flow through a composite plane wall between two fluids, or between two known surface temperatures: each layer's conduction resistance, each film's convection resistance, the overall U-value, the heat flow and flux, and every surface and interface temperature.

WHAT IT CALCULATES	RELATION
Resistance of each surface film	$R = 1/(h \cdot A)$
Resistance of each conducting layer	$R_i = t_i/(k_i \cdot A)$
Total resistance, hot side first	$R_{tot} = \Sigma R$
Heat flow through the wall	$Q = (T_1 - T_2)/R_{tot}$
Heat flux and overall U-value	$q = Q/A$; $U = 1/(R_{tot} \cdot A)$
Surface and interface temperatures, stepping down from T_1 across each resistance in turn	$T_{next} = T_{previous} - Q \cdot R$

Basis: The plane wall: thermal resistance and the composite wall[1](§3.1); Generalized thermal resistance networks (multilayer plane walls)[2](§3-3); Thermal resistance of building components (method)[3](§6).

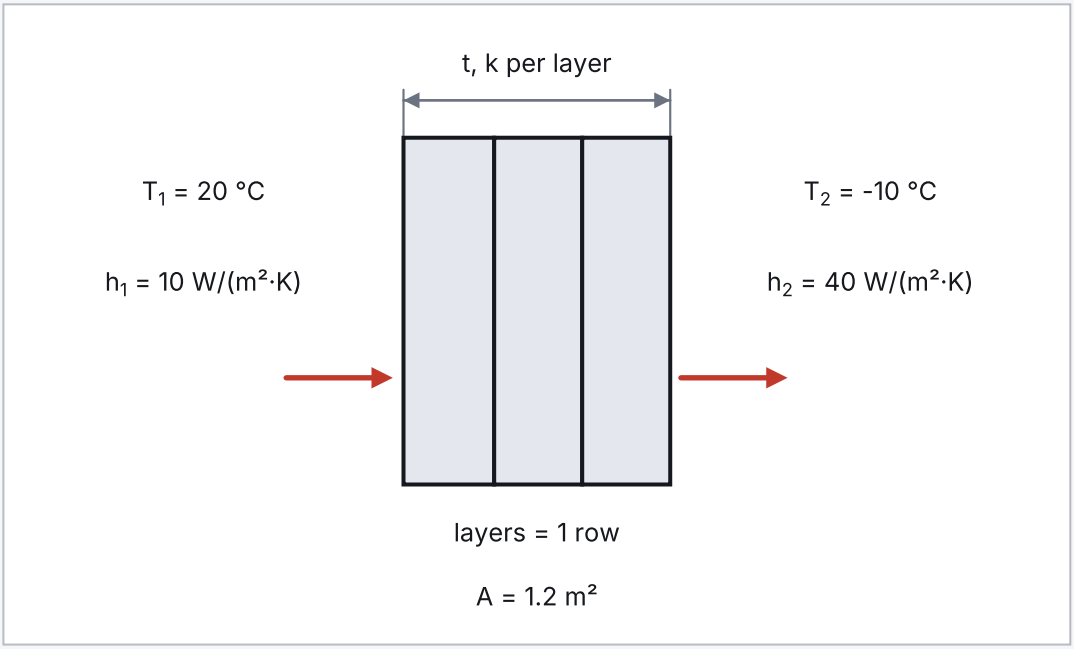


Figure 1 — A composite plane wall, side 1 left: a fluid film, the layers as listed (three drawn), a film on side 2. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
T_1	Side 1 temperature (the fluid's, or the surface's when no film is given)	20	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_1	Side 1 film coefficient (omit when T_1 is the surface temperature)	10	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
layers	Conducting layers, side 1 first	1 row(s)		drawing: DRG-GL-801 rev A	any
h_2	Side 2 film coefficient (omit when T_2 is the surface temperature)	40	W/(m²·K)	spec: SPEC-BS-04 design conditions	2 … 10000 W/(m²·K)
T_2	Side 2 temperature (the fluid's, or the surface's when no film is given)	-10	°C	spec: SPEC-BS-04 design conditions	-60 … 400 °C
A	Wall area normal to the heat flow	1.2	m²	drawing: DRG-GL-801 rev A	0.001 … 10000 m²

TABLE ROW	VALUES
layers[1]	name = glass; t = 8 mm; k = 0.78 W/(m·K)

3. Assumptions

A1. Steady, one-dimensional conduction through plane layers of constant conductivity, with no internal heat generation. (Thermal-circuit idealisation: equal layer areas, no edge losses.)
A2. Layers are in perfect thermal contact; each film coefficient is constant over its surface. (A contact resistance can be entered as a thin layer of equivalent thickness and conductivity.)

4. Working

[6] $R_{h1} = 1 / (h_1 \cdot A)$ [1](§3.1)
= 1 / (10 W/(m²·K) × 1.2 m²)
= 0.083333 K/W
Side 1 film resistance

[7] $R_1 = t_1 / (k_1 \cdot A)$ [1](§3.1)
= 8 mm / (0.78 W/(m·K) × 1.2 m²)
= 0.008547 K/W
Conduction resistance of glass

[8] $R_{h2} = 1 / (h_2 \cdot A)$ [1](§3.1)
= 1 / (40 W/(m²·K) × 1.2 m²)
= 0.020833 K/W
Side 2 film resistance

[9] $R_{tot} = \Sigma R$ [2](§3-3)
= 0.083333 + 0.008547 + 0.020833 K/W
= 0.11271 K/W
Total thermal resistance

[10] $Q = (T_1 - T_2) / R_{tot}$
= (20 degC − (-10 degC)) / 0.11271 K/W
= 266.16 W
Heat flow from side 1 to side 2

[11] q = Q / A
= 266.16 W / 1.2 m²
= 221.8 W/m²
Heat flux

[12] $U = 1 / (R_{tot} * A)$
$= 1 / (0.11271\ K/W \times 1.2\ m^2)$
$= 7.3934\ W/(m^2 \cdot K)$
Overall heat transfer coefficient
[13] $T_{s0} = T_1 - Q \cdot R_{h1}$
$= 20\ °C - 266.16\ W \times 0.083333\ K/W$
$= -2.1801\ degC$
Side 1 surface
[14] $T_{s1} = T_{s0} - Q \cdot R_1$
$= -2.1801\ °C - 266.16\ W \times 0.008547\ K/W$
$= -4.455\ degC$
Side 2 surface

5. Results

SYMBOL	VALUE	DESCRIPTION
Q	266.2 W	Heat flow
q	221.8 W/m²	Heat flux
U	7.393 W/(m²·K)	Overall heat transfer coefficient
R_{tot}	0.1127 K/W	Total thermal resistance
R_{h1}	0.08333 K/W	Side 1 film resistance
R_1	0.008547 K/W	Layer 1 resistance
R_{h2}	0.02083 K/W	Side 2 film resistance
T_{s0}	-2.180 °C	Side 1 surface temperature
T_{s1}	-4.455 °C	Side 2 surface temperature

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

Template bs.plane-wall@1.0.1 Content hash d3270684be534c78 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Bergman TL, Lavine AS, Incropera FP, DeWitt DP. Fundamentals of Heat and Mass Transfer. 7th ed. Wiley; 2011.
2. Çengel YA, Ghajar AJ. Heat and Mass Transfer: Fundamentals and Applications. 5th ed. McGraw-Hill Education; 2015.
3. British Standards Institution. Building Components and Building Elements—Thermal Resistance and Thermal Transmittance—Calculation Methods. BS EN ISO 6946. BSI.